

Use number skills in algebraic and graphical contexts

Apply number skills to simplify expressions, solve equations and interpret graphs.

Step 56

★ Today I am learning to use number skills in algebraic and graphical contexts.

1. Simplify Algebraic Expressions

Simplify each expression.

- | | |
|------------------------------|-----------------------------------|
| a) $3x + 4x - 7x =$ _____ | e) $2(x + 5) - 3x =$ _____ |
| b) $5a - 2a + 3a =$ _____ | f) $4(2y - 1) + 3y =$ _____ |
| c) $6y + 8 - 2y - 3 =$ _____ | g) $5(2a - 3) - 2(a + 4) =$ _____ |
| d) $7b - 3(b + 2) =$ _____ | h) $3(x - 2) + 2(4 - x) =$ _____ |

Remember:

- Collect like terms.
- Multiply out brackets first.
- Simplify each step.

2. Solve Linear Equations

Solve each equation.

- | | | |
|--------------------------|--------------------------|------------------------------|
| a) $3x + 5 = 20$ | e) $2(x + 5) = 24$ | i) $3x + 2 = 5x - 6$ |
| b) $2x - 7 = 11$ | f) $3(x - 4) = 15$ | j) $2(3x - 1) = 4x + 8$ |
| c) $5x = 35$ | g) $\frac{x + 7}{2} = 9$ | k) $\frac{x - 3}{3} + 2 = 7$ |
| d) $\frac{x}{4} + 3 = 8$ | h) $5 - 2x = -11$ | l) $4 - (x + 2) = 2(x - 5)$ |

Think about it!

- Do the opposite operation to both sides.
- Keep the equation balanced.
- Check your solution.

3. Substitute Numbers

Substitute the given value(s) into the expressions and simplify.

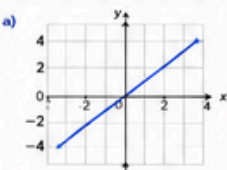
- | | |
|-----------------------------------|--|
| a) $2x + 3$, when $x = 5$ | e) $2(x + 1)(x - 2)$, when $x = 4$ |
| b) $3a - 4$, when $a = -2$ | f) $y(y - 5) + 3y$, when $y = -2$ |
| c) $4b^2 - b$, when $b = 3$ | g) $(2m^2 - m + 1) - (m^2 + 3)$, when $m = 2$ |
| d) $x^2 + 2x - 1$, when $x = -3$ | h) $3p(p + 2) - (p^2 - 4)$, when $p = -1$ |

Remember:

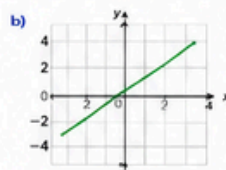
- Substitute carefully.
- Follow the order of operations.
- Simplify your answer.

4. Coordinate Geometry

Find the gradient and y-intercept of each line. Then write the equation in the form $y = mx + c$.



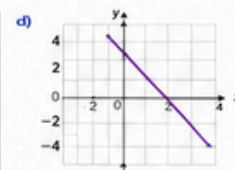
Gradient: _____ y-intercept: _____
Equation: _____



Gradient: _____ y-intercept: _____
Equation: _____



Gradient: _____ y-intercept: _____
Equation: _____



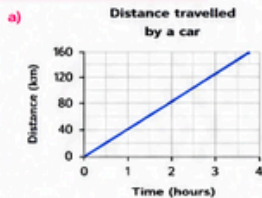
Gradient: _____ y-intercept: _____
Equation: _____

Think about it!

- Gradient (m): rise/run
- y-intercept (c) is where the line crosses the y-axis.
- Use two clear points from the graph.

5. Interpreting Graphs

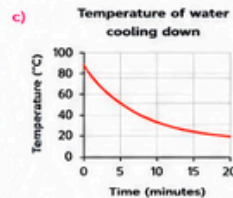
Use the given graphs to answer the questions.



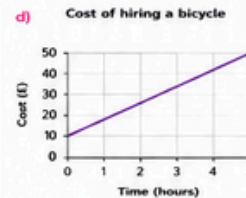
- How far has the car travelled in 4 hours?
- What is the distance after 2.5 hours?



- How much was in the account initially?
- How much is in the account after 5 months?



- What was the initial temperature?
- What is the temperature after 15 minutes?



- What is the fixed charge?
- What is the cost for 3 hours?

Remember:

- Read the scales carefully.
- Look at the axes labels and units.
- Use the graph to estimate or find exact values.

6. Real-World Problems (Algebraic)

Translate each situation into an algebraic expression or equation and solve.

- | | | | |
|--|---|---|--|
| a) The sum of three times a number and 7 is 22. What is the number?

Answer: _____ | b) Twice a number minus 5 is equal to three times the number plus 4. What is the number?

Answer: _____ | c) A number increased by 15 is two times the number. What is the number?

Answer: _____ | d) The perimeter of a rectangle is 54 cm. Its length is 3 cm more than twice its width. Find the length and width.

Answer: length = _____ width = _____ |
|--|---|---|--|

Think about it!

- Read the problem carefully.
- Let the unknown be x .
- Form an equation.
- Solve and check.

7. Rate of Change

Find the rate of change for each situation.

- | | | | |
|---|--|---|--|
| a) A car travels 180 km in 3 hours. What is the average speed in km/h?

Answer: _____ | b) The number of bacteria increases from 500 to 2,000 in 6 hours. Find the rate of increase per hour.

Answer: _____ | c) A savings account increases from £150 to £195 in 5 months. Find the rate of increase per month.

Answer: _____ | d) The price of a laptop decreases from £600 to £450 in 10 months. Find the rate of decrease per month.

Answer: _____ |
|---|--|---|--|

Remember:

- Rate of change = $\frac{\text{change in amount}}{\text{change in time}}$
- Include units.

My Learning Check

Circle how confident you feel.



I can do it!



Almost there!



I need more practice.



I need help.

Parent Observation (optional)

Please tick the boxes that apply.

- | | |
|--|--|
| ☐ Understands concepts | ☐ Explains thinking clearly |
| ☐ Uses number skills accurately | ☐ Needs more practice |
| ☐ Solves problems confidently | |

Notes: _____

Today I learned...

1 thing I did well: _____

1 thing I will improve: _____